

SE LAB-1

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Problem Statement-43:

Feature Flag & Dynamic Config Manager

1) Requirements Table

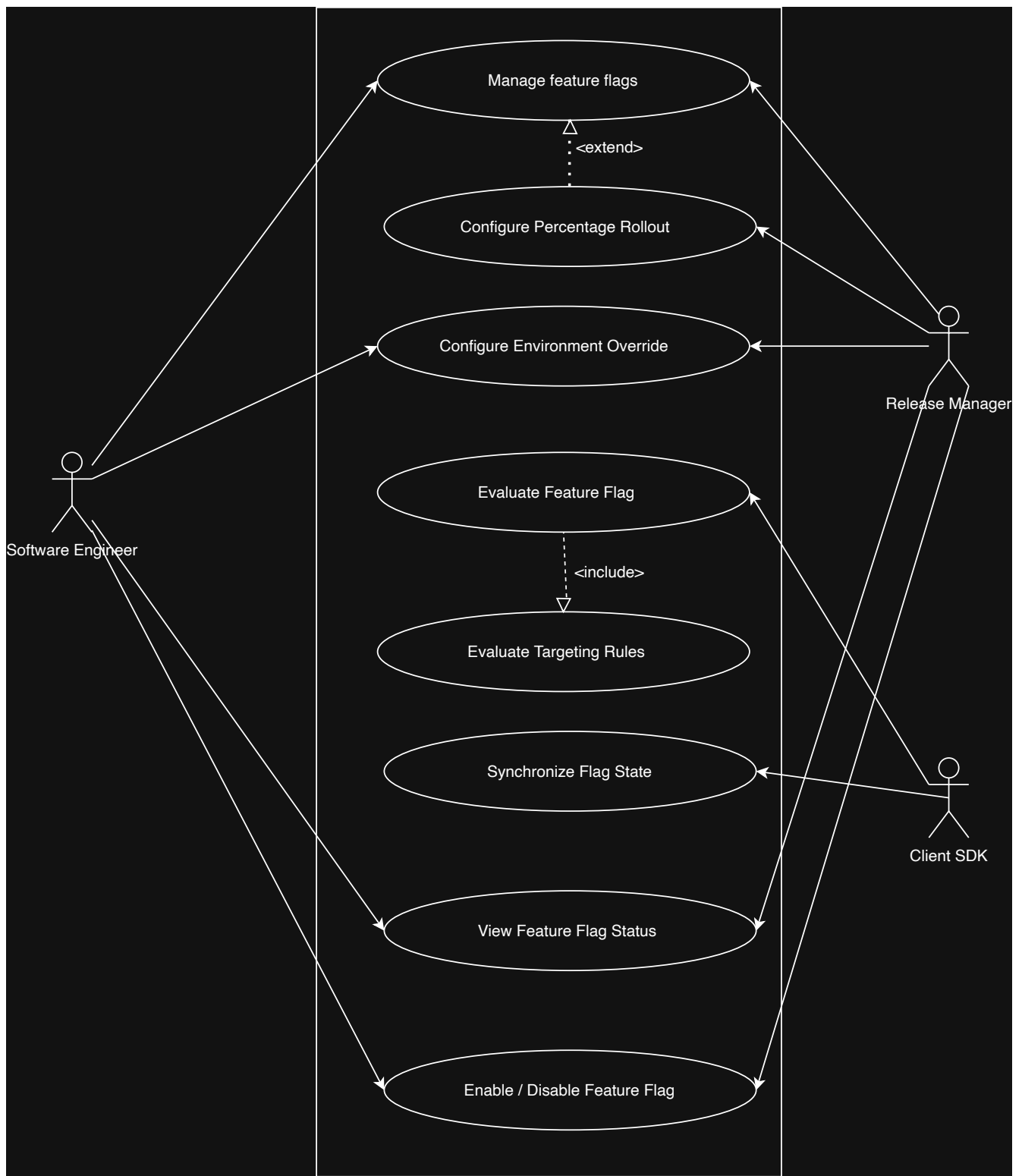
REQ ID	TYPE	DESCRIPTION	PRIORITY	ACCEPTANCE CRITERIA	RATIONALE
FR-001	Functional	The system shall evaluate user targeting rules such as User ID hash and Beta cohort and return the appropriate boolean feature flag state.	High	A user belonging to a 10% rollout cohort receives the enabled flag.	Core functionality for controlling feature availability.
FR-002	Functional	The system shall allow a Release Manager to configure percentage-based feature rollouts for selected user cohorts.	High	When a 10% rollout is configured, users selected according to the rollout rule receive the enabled flag.	Enables gradual and controlled feature releases.
FR-003	Functional	The system shall allow a Software Engineer or Release Manager to configure environment-specific feature flag overrides.	High	A flag configured differently for Development and Production returns the corresponding environment-specific value.	Allows features and configurations to differ across environments.

REQ ID	TYPE	DESCRIPTION	PRIORITY	ACCEPTANCE CRITERIA	RATIONALE
FR-004	Functional	The system shall synchronize updated feature flag states with connected client SDKs in real time.	High	After a flag is changed, a connected SDK receives the updated flag state without requiring a manual application restart.	Ensures applications use the latest configuration.
FR-005	Functional	The system shall allow authorized users to create, update, enable, disable, and view feature flags.	Medium	An authorized user can create and modify a flag and the updated state is displayed when queried.	Provides the basic management functionality required by the platform.
NFR-001	Non-Functional – Performance & Security	The feature flag evaluation API shall execute in under 15 ms to minimize overhead when integrated into client critical paths.	High	Benchmarking confirms API evaluation latency is below 15 ms under the specified test load.	Ensures feature-flag evaluation does not significantly affect application performance.
NFR-002	Non-Functional – Reliability	The system shall maintain correct feature flag states during client SDK synchronization and configuration updates.	Medium	During a synchronisation test, all connected clients receive the latest valid flag state without inconsistent values.	Prevents applications from using outdated or inconsistent configurations.

2) UML USE-CASE DIAGRAM

ACTION:

- Software Engineer
- Release Manager
- Client SDK



3) Use Case Flow

Preconditions:

1. The feature flag exists in the system.
2. The Client SDK is connected to the Feature Flag Manager.
3. The feature flag has valid targeting rules.
4. The user/request contains the required identification information.

Postconditions:

5. The system returns the evaluated feature flag state.
6. The Client SDK receives either true/enabled or false/disabled.
7. The system returns an appropriate failure/default response if the flag cannot be evaluated.

Main Success Scenario:

1. Client SDK sends a feature flag evaluation request to the system.
2. System identifies the requested feature flag.
3. System retrieves the flag configuration and targeting rules.
4. System evaluates the user's attributes against the configured targeting rules.
5. System determines whether the user belongs to the configured rollout cohort.
6. System checks any applicable environment override.
7. System determines the final boolean feature flag state.
8. System returns the feature flag state to the Client SDK.
9. Client SDK applies the returned state and enables or disables the feature accordingly.
10. Use case ends successfully.

Alternative Scenario:

User Does Not Match Targeting Rule

5a. The user's attributes do not satisfy the configured targeting rule.

5a1. System evaluates the user as outside the enabled cohort.

5a2. System determines the feature flag state as disabled.

5a3. System returns false to the Client SDK.

5a4. Client SDK keeps the feature disabled.

5a5. Use case ends.